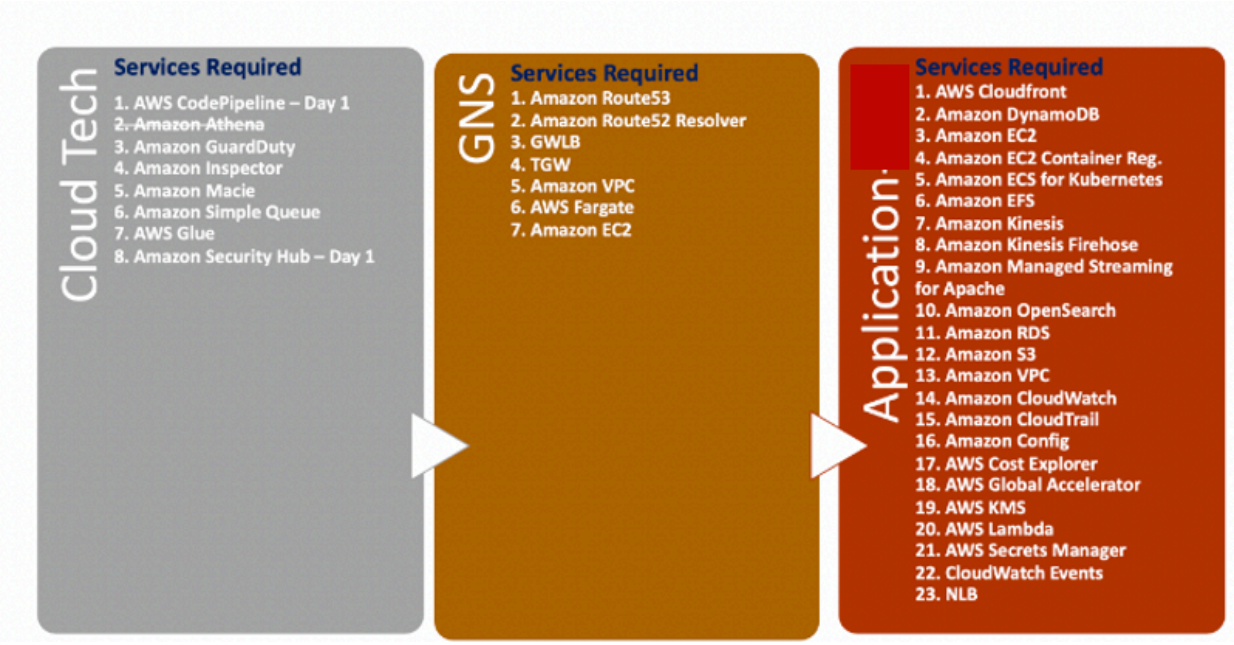
New Region Expansion - HYD / ap-south-2 [Hyderabad] working document

Region Data - India (Hyderabad) - HYD

Region name	ap-south-2
Availability Zones:	ap-south-2a
	ap-south-2b
	ap-south-2c
FS target need-by date (Applications):	q4` 25
FS target need-by date (Platform):	q4` 25

Region Readiness Summary



GNS - Route 53(Route 52 Resolver), Ec2 (GWLB), Direct Connect (TGW), VPC, Fargate

Latest Services per Coud Tech :

Service	Available?	Day 1 Blocker?	Endpoints available in all AZs?	Required for?	Team	Date expected	Comments	Difference in Service Operations
AWS CodePipeline	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon DynamoDB	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon EC2	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon Elastic Container Service for Kubernetes	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon GuardDuty	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon Route53	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon Route53 Resolver	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon Simple Queue	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon S3	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon VPC	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon Cloud Watch	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS CloudTrail	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Config	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS KMS	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Lambda	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Security Hub	Yes	No	Yes	Platform	Cloud Tech	Available		
Amazon VPC - Transit Gateway (TGW)	Yes	No	Yes	Platform	GNS	Available		
AWS CloudFormation	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS IAM	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Private Link	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Secrets Manager	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Security Token Service	Yes	No	Yes	Platform	Cloud Tech	Available	sts.ap-south-2.amazonaws.com	
AWS Step Functions	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Support	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS Systems Manager	Yes	No	Yes	Platform	Cloud Tech	Available		
AWS CloudWatch Events	Yes	No	Yes	Platform	Cloud Tech	Available		

Additional Documentation / References

- Public Information / Resources
  - AWS list of regions: https://aws.amazon.com/about-aws/global-infrastructure/regions_az
  - AWS list of Services by Region: https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services
- EXAMPLE: Business Justification Reference

Questions for BU planning to migrate/expand in specific region:

1. Has the BU unit opened Radar ticket to Cloud Tech. If not, please prioritize this.
2. Application level checklist

1. Pre-Expansion Assessment
- [ ] Business justification for expansion
- [ ] Target region selection criteria met
- Latency requirements
- Data sovereignty requirements
- Cost analysis

- Service availability in target region
- [ ] Compliance and regulatory requirements review
- [ ] Budget approval

2. Infrastructure Planning
- [ ] VPC Design
- CIDR blocks
- Subnet allocation
- Availability Zones utilization
- [ ] Network Connectivity
- [ ] VPC peering requirements
- [ ] Transit Gateway configuration
- [ ] Direct Connect needs
- [ ] Route53 strategy

3. Service Migration Checklist
Core Services
- [ ] EC2 instances
- AMI replication
- Instance types availability
- [ ] RDS/Database strategy
- Read replicas
- Multi-region setup
- [ ] S3 bucket replication
- [ ] IAM policies and roles

Supporting Services
- [ ] CloudFront distribution updates
- [ ] Load Balancer configuration
- [ ] Auto Scaling groups
- [ ] CloudWatch monitoring
- [ ] Lambda functions deployment

4. Data Management
- [ ] Data replication strategy
- [ ] Backup procedures
- [ ] Data sync mechanisms
- [ ] Latency considerations

5. Security Implementation
- [ ] Security groups configuration
- [ ] NACLs setup
- [ ] KMS keys in new region
- [ ] WAF rules replication
- [ ] Certificate management

6. Deployment Strategy

- [ ] Blue-Green deployment plan
- [ ] Rollback procedures
- [ ] Testing strategy
- Performance testing
- Failover testing
- Integration testing

7. Operations
- [ ] Monitoring setup
- [ ] Alerting configuration
- [ ] Runbooks update
- [ ] Documentation update
- [ ] Support team training

8. Cost Management
- [ ] Budget alerts setup
- [ ] Cost allocation tags
- [ ] Resource optimization plan
- [ ] Reserved capacity planning

9. Post-Deployment
- [ ] Performance baseline establishment
- [ ] Disaster recovery testing
- [ ] Documentation finalization
- [ ] Operational handover